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Major Project Report on AI-Powered Image Text Translation & Editing

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Certificate of Originality

This is to certify that the project entitled "**Image-Based OCR Translation and Text Overlay System using FastAPI**" submitted by B Bhakta Varun, Enrolment No. 01719011621, is a record of original work carried out by the student under my supervision. This work has not been submitted elsewhere for any other degree or diploma.

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Abstract

In an increasingly interconnected world, language remains one of the most significant barriers to global communication and access to information. With the rise of globalization, people often encounter foreign-language text embedded in real-world environments such as signboards, scanned documents, manuals, educational resources, or street directions. Translating such content manually or using traditional tools can be tedious, error-prone, and inefficient, especially when text appears in image formats rather than editable documents.

This project presents an AI-powered image translation system that integrates Optical Character Recognition (OCR) with machine translation and real-time visualization. The core objective is to enable users to seamlessly detect, translate, and understand foreign language text within static images. The system is designed using two main technologies: EasyOCR for text detection and FastAPI for web-based deployment and image processing services.

The workflow begins with the user uploading an image containing foreign text, specifically Japanese or English. The OCR engine scans the image and identifies text regions along with their spatial coordinates. These regions are then extracted and passed to a translation service powered by the MyMemory Translation API, which automatically translates the text into the user's preferred language.

One of the distinguishing features of the system is its intelligent overlay mechanism. Once the translation is obtained, the system calculates a suitable font size for the translated text based on the height of the original bounding box using a scaling function (e.g., $\text{Font}(h) = \max(10, \text{int}(0.8 \times h))$). This ensures that the overlay text fits proportionally and remains visually readable. To further enhance clarity, the system draws a white background rectangle beneath the translated text to mask the original content. This prevents any overlap or confusion caused by the original and translated texts appearing simultaneously.

The entire system is hosted on a lightweight API-based architecture built with FastAPI. It allows users to access the service via RESTful endpoints, making it scalable and easy to integrate into existing applications or mobile platforms. The modular design includes separate components for OCR processing, translation, image annotation, and API routing—making it highly extensible for future improvements such as supporting more languages or integrating additional OCR models like Tesseract or PaddleOCR.

From a usability perspective, this tool caters to a wide range of practical use cases. Travelers can instantly translate street signs or notices in unfamiliar languages, educators can digitize and localize foreign study materials, and professionals can process international documents without needing separate OCR and translation tools. The system runs efficiently on standard CPUs, requiring no specialized hardware like GPUs, which makes it accessible for everyday users and institutions with limited resources.

In terms of performance, initial testing shows promising results: the OCR accuracy reaches approximately 85%, and the average translation latency is around one second per text block. The system can dynamically adjust font sizes between 10 px to 40 px, ensuring optimal visibility across different image types and resolutions.

While comparable mobile apps like Google Translate offer similar functionality, they often use proprietary systems, have fixed font overlays, or lack custom deployment options. In contrast, the proposed system is open, modifiable, and easy to deploy on any FastAPI-supported

backend, making it a strong choice for developers, researchers, and institutions seeking customizable solutions.

In conclusion, this project contributes a functional and adaptable pipeline that bridges OCR, translation, and visualization into a single, easy-to-use application. It represents a step toward making multilingual communication more accessible and efficient in a visually driven digital age.

Introduction

In the era of globalization, the ability to understand and interpret content across multiple languages has become an essential skill in both professional and everyday contexts. From travel and tourism to education, international business, and global communication, people frequently encounter information in languages they do not understand. Text embedded in images, such as road signs, restaurant menus, scanned documents, advertisements, and educational posters, often poses a significant challenge for non-native speakers. Unlike typed digital text, these image-based texts cannot be easily copied, translated, or edited using conventional tools, creating a barrier to accessibility.

To address this, technologies such as Optical Character Recognition (OCR) and machine translation have emerged as vital tools. OCR enables the extraction of textual data from images by identifying and converting characters into machine-encoded text. Meanwhile, machine translation technologies can automatically translate one language into another with increasing accuracy, powered by neural networks and vast multilingual datasets.

Although both technologies have matured independently, their integration into a seamless, real-time system remains relatively underexplored in practical applications. Existing tools typically handle OCR and translation as separate processes, requiring users to use multiple applications or manually copy extracted text for translation. Moreover, visual representation of the translated text—such as overlaying it within the image context—is rarely considered in these solutions.

This project proposes a web-based, AI-powered system that combines OCR and machine translation into a single, unified workflow. The system is designed to be lightweight, modular, and accessible through a FastAPI-powered web interface, allowing users to upload images, detect foreign text, translate it to their preferred language, and receive a visually-enhanced image with translated text overlays—all in real time.

The implementation leverages EasyOCR, a robust and open-source OCR engine known for supporting multiple languages, including Japanese and English. The choice of these two languages allows for testing across a linguistically diverse spectrum—one based on Latin characters and the other using logographic scripts. After the text is detected, the MyMemory Translation API is used to perform the translation. The processed results are then rendered visually using the Python Imaging Library (PIL), where bounding boxes are drawn around the original text regions, a white background is inserted for contrast, and the translated text is overlaid with dynamically adjusted font size for readability.

One of the key motivations behind this project is usability in multilingual real-world environments. For instance, a traveler in Japan trying to read a signboard, or a student working with a scanned document in a foreign language, can benefit greatly from such a tool. Traditional OCR software may extract the text, and separate translation tools can provide meanings, but

rarely does a system integrate both steps while also returning a clearly readable version of the translated content within its original visual context.

This project also introduces an API-first architecture, which enables extensibility and integration into larger systems such as mobile apps, enterprise tools, or educational platforms. Developers can interact with the backend using well-defined endpoints to submit images and retrieve processed results, making it easy to incorporate into broader workflows.

Furthermore, the system is optimized for resource-constrained environments and does not rely on GPU acceleration or cloud-based computation. It runs locally on standard CPU-based systems, ensuring that institutions or users with limited computational infrastructure can still benefit from its functionality.

In summary, this project aims to create an intelligent, responsive, and user-friendly solution that enhances cross-linguistic communication through the seamless fusion of OCR, machine translation, and image visualization. The resulting tool not only addresses a common pain point in global communication but also serves as a stepping stone toward more intelligent, accessible, and human-centered language processing systems.

Problem Statement

In an age marked by globalization and technological advancement, the exchange of information across languages and cultures has become commonplace. However, for individuals who do not speak or read a particular language, accessing information embedded in foreign scripts remains a significant obstacle—especially when that information is embedded in non-editable image formats such as photographs, scanned documents, posters, or signage. For non-native speakers, reading and understanding such content is often frustrating, time-consuming, and practically inaccessible without external help.

A common scenario can be seen in international travel: tourists frequently encounter foreign-language signs in airports, train stations, restaurants, or on the street. Similarly, students and professionals often deal with scanned academic materials, research papers, user manuals, or business documents in unfamiliar languages. In such contexts, manual translation—which requires typing out or describing the text for translation—is not only tedious but also error-prone and inefficient. The inability to interact directly with the text in its original visual context hinders understanding and delays critical decision-making.

To solve this problem, technologies like Optical Character Recognition (OCR) and machine translation have been developed and widely adopted. OCR enables machines to detect and extract text from images, converting static characters into editable, searchable text. Machine translation, on the other hand, provides fast and reasonably accurate translation across various language pairs. However, while both technologies have matured independently, their integration into a single, seamless workflow is still lacking in most mainstream tools.

Most existing OCR applications are limited to text extraction and stop short of providing built-in translation features. Even when translation is available—such as in the Google Translate mobile app—it often lacks user-friendly visualization, such as the ability to cleanly replace the original text within the image or offer customization in presentation. This often results in cluttered visuals or translations that are difficult to read due to overlapping characters or inconsistent font sizes. Users may also be restricted to proprietary platforms without the ability to deploy or modify the system based on specific

needs, making them unsuitable for educational institutions, researchers, or developers looking for customizable solutions.

Moreover, these tools generally do not account for visual clarity. Original text and translated content often overlap, making it harder for users to decipher meaning. In many cases, the translated text is simply presented below the image or in a separate field, losing the spatial and contextual connection to the original image. This disconnection can diminish the user's understanding, especially in images with multiple text regions such as charts, infographics, or labeled diagrams.

This project aims to bridge these gaps by developing an AI-powered tool that not only extracts and translates text from images but also reinserts the translated content directly into the original image in a visually clear and readable manner. It employs EasyOCR to detect Japanese and English text and uses the MyMemory Translation API to convert the detected text into the user's preferred language. Crucially, it introduces a feature-rich overlay mechanism: bounding boxes are drawn around the detected text, a white background is inserted to mask the original text, and the translated text is overlaid with dynamically adjusted font sizes to ensure proportional and legible display.

The system also features a modular FastAPI-based backend, allowing for RESTful API interactions. This makes the tool accessible for integration into broader platforms such as web applications, mobile services, or classroom tools. Users can upload images via an endpoint and receive a processed image with all translations embedded in-place—offering a seamless and intuitive experience.

In conclusion, the core problem this project addresses is the lack of an end-to-end, user-friendly system that can detect, translate, and visually integrate translated text into images. By solving this challenge, the system empowers users to understand and interact with multilingual image content with ease, opening the door to more inclusive and accessible communication in a globalized world.

Literature Survey

The combination of Optical Character Recognition (OCR) and machine translation has the potential to transform how users interact with multilingual content in image formats. Over the past decade, significant strides have been made in both fields individually. However, there is still a noticeable lack of integrated systems that combine robust OCR, high-quality translation, and user-friendly visualization into a single automated pipeline.

Tesseract OCR, developed by Google, is one of the most widely used open-source OCR engines. It supports over 100 languages and delivers reliable results for clean, well-structured documents. Despite its strengths, Tesseract struggles with more complex scenarios such as handwritten or low-quality text, and its output often requires substantial preprocessing. On the other hand, EasyOCR, a newer deep learning-based OCR tool, provides superior performance on a wide variety of fonts and image types. It supports over 80 languages and excels at recognizing irregular or artistic fonts, making it suitable for real-world applications like signage or scanned books.

In the realm of machine translation, services such as the MyMemory API and Google Translate API have been extensively utilized for both academic and commercial purposes. These APIs leverage extensive language corpora and deep learning techniques to provide near real-time translation with high accuracy. While powerful, these services are typically accessed in isolation and often lack seamless integration with image-processing systems.

Despite the availability of these tools, few existing solutions offer a unified, end-to-end system that extracts text from images, translates it, and visually overlays the translated content within the same image in an intelligent and readable manner. Most applications either extract text and provide

translation separately or display translations outside the image context—failing to deliver an immersive or intuitive experience. Moreover, few offer users any control over visualization aspects like font size, overlay clarity, or image formatting.

This project addresses those gaps by proposing a modern, lightweight, and modular solution that integrates OCR, translation, and image rendering using an API-first architecture. The design emphasizes real-time interactivity, clean visualization, and modularity, enabling developers to adapt or scale the system as needed. Unlike static tools, this system dynamically adjusts to each input image, making it highly adaptable to different languages, fonts, and layout complexities.

Name and Description of Major Modules

To achieve a clean, scalable architecture, the system is divided into four major functional modules:

1. OCR Module

This module is responsible for detecting and extracting textual content from images. It uses EasyOCR, which supports both Japanese and English—languages with distinct scripts and structures. EasyOCR processes the uploaded image and returns a list of detected text regions along with their bounding box coordinates. This information is critical for both translation and visual overlay. The OCR module handles preprocessing tasks such as grayscale conversion and resizing to optimize detection performance.

2. Translation Module

Once text is extracted, it is passed to the Translation Module, which interfaces with the MyMemory Translation API. This API supports a wide range of language pairs and is particularly useful for building proof-of-concept systems due to its simplicity and free-tier access. The module ensures that text from each bounding box is translated individually and that language pair selection is flexible based on user input. Error handling mechanisms are also included to deal with translation limits or network failures.

3. Overlay Module

This is a critical component that focuses on user-friendly visualization. It takes the original image and the translated text, calculates the appropriate font size using the height of the bounding box ($\text{Font}(h) = \max(10, \text{int}(0.8 \times h))$), and draws a white rectangle over the original text to enhance readability. The translated text is then overlaid using the Python Imaging Library (PIL). This module ensures that translations are clear, legible, and do not overlap awkwardly with original image elements, making the final output visually polished and useful for real-world use.

4. API Layer

The system is wrapped in a FastAPI-based backend that provides simple and efficient endpoints for interaction. Users can upload images, configure translation preferences, and retrieve processed images via RESTful endpoints. FastAPI ensures high performance, auto-generated documentation, and easy integration with frontend clients or other services. This modular structure allows the system to be extended with additional features, such as support for more languages, batch processing, or cloud-based storage.

Design Flow

The design of the proposed system follows a modular, pipeline-based approach, ensuring clear separation of responsibilities, ease of maintenance, and scalability. This section elaborates on the step-by-step process of the system using visual representations such as flow charts, data flow diagrams, and entity-relationship diagrams to communicate how data is processed and transferred within the system.

a. Flow Chart Design

The flow chart represents the high-level execution flow of the system from start to end. It outlines the user's interaction with the system and the internal processing steps the image undergoes. This ensures a clear understanding of how the input (image) is transformed into the output (translated and processed image).

Flow Chart Steps:

Start: The process begins when a user initiates a request to the system.

Upload Image: The user uploads an image containing text in either Japanese or English. This is handled through the FastAPI interface.

Extract Text Using OCR: The system uses EasyOCR to detect text regions and recognize the textual content from the image. The output includes both the text and its bounding box coordinates.

Translate Text: The extracted text is passed to the MyMemory Translation API, which converts it into the user's target language.

Draw Bounding Boxes & Overlay Translated Text with White Background: The system calculates appropriate font sizes based on bounding box height, overlays a white rectangle to obscure the original text, and writes the translated text using PIL.

Return Processed Image: The final image, containing cleanly overlaid translated text, is returned to the user.

End: The operation concludes.

This sequential pipeline allows for real-time, efficient processing of multilingual content embedded in images.

b. Data Flow Diagram (DFD)

The Data Flow Diagram describes how data moves between different components of the system. It focuses on the flow of information rather than the specific sequence of operations, offering insight into the interactions between user input, system processing units, and output delivery.

DFD Description:

User → FastAPI Endpoint: The user initiates the request by uploading an image via an HTTP POST request. The image and language preferences are sent to the backend.

FastAPI Endpoint → OCR Module: The backend routes the uploaded image to the OCR module. EasyOCR processes the image to detect and extract text and their respective locations.

OCR Module → Translation API: Extracted text segments are sent to the MyMemory Translation API. Each piece of text is translated into the preferred language.

Translation API → Image Processing Module: The translated text, along with bounding box coordinates, is sent to the image processor.

Image Processing Module → Response with Translated Image: The image is modified to include

translated overlays and is returned to the user via the same FastAPI endpoint.

This DFD structure ensures a well-organized, maintainable architecture where each module performs a dedicated role.

c. E-R Diagram (Simplified)

The Entity-Relationship Diagram models the data relationships involved in the system. While traditional E-R diagrams are used for database design, a simplified version helps visualize how major entities in the system are related to one another during processing.

Simplified E-R Relationships:

[User] --uploads--> [Image]: The user interacts with the system by uploading an image file.

[Image] --contains--> [TextRegion]: The image consists of one or more regions containing text that needs to be recognized and translated.

[TextRegion] --translated by--> [Translation API]: Each detected text region is sent to the translation service, which returns translated text.

[Translation API] --produces--> [TranslatedText]: The API provides the translated output, which will then be inserted into the original image.

This model demonstrates the chain of relationships from user input to final output while mapping out key processing entities.

Methodology Adopted

To implement the proposed multilingual image translation and overlay system, a well-structured algorithm is formulated, accompanied by a mathematical model to define its functional behavior. This section also outlines the simulation environment used for development and testing.

a. Algorithm Formulated

The algorithm for the system follows a modular and sequential pipeline, ensuring clarity, maintainability, and performance optimization. It is designed to process any image containing Japanese or English text and return a translated version with visually overlaid text for better readability.

Step-by-Step Algorithm:

Accept Input Image from User

The user uploads an image through a FastAPI endpoint. The image is temporarily stored on the server for processing.

Convert Image to Format Suitable for OCR

The uploaded image is preprocessed, if necessary, by converting it into a format compatible with EasyOCR. This may include resizing, grayscale conversion, or noise reduction to improve text detection accuracy.

Detect Text Regions Using EasyOCR

The EasyOCR library processes the image and returns a list of detected text instances. Each instance includes the bounding box coordinates, the recognized text, and the confidence score.

Iterate Over Each Detected Region:

For every detected region, the following sub-steps are performed:

Extract Text and Bounding Box

The system extracts the textual content and its spatial bounding box coordinates.

Send Text to MyMemory API for Translation

The extracted text is sent via an HTTP request to the MyMemory Translation API. The translated text is returned in the target language specified by the user.

Calculate Appropriate Font Size Based on Bounding Box Height

Font size is dynamically computed as a function of the bounding box height to ensure the translated text fits naturally in the original space.

Draw White Rectangle Over Original Text

A white rectangular mask is drawn over the original text region to obscure it and provide contrast for the new text overlay.

Overlay Translated Text Using PIL (Python Imaging Library)

The translated text is rendered on top of the white rectangle with the computed font size, ensuring legibility and preserving the spatial context.

Save and Return the Processed Image

The final image, containing all translated overlays, is saved and returned to the user through the API endpoint.

This algorithm ensures a fully automated, real-time translation system that maintains the visual layout of the original image while enhancing its accessibility.

b. Mathematical Modelling

To formalize the processing logic of the system, we define the following variables and functions:

Let $T = \{t_1, t_2, \dots, t_n\}$ represent the set of n texts detected in the image.

Let $B = \{b_1, b_2, \dots, b_n\}$ represent the corresponding set of bounding boxes for each text.

Define the font size function $F(h)$ based on the height h of the bounding box:

$$F(h) = \max(10, \text{int}(0.8 \times h))$$

Let $\tau(t_i)$ be the translation function that maps the original text t_i to its translated equivalent τ_i in the target language.

Thus, for each pair (t_i, b_i) , the final image is generated by overlaying $\tau(t_i)$ using $F(b_i.\text{height})$ on the masked region defined by b_i .

This model ensures proportional text scaling and accurate placement for each translated text element.

c. Simulation Environment

To develop and validate the system, the following tools and technologies were utilized:

Programming Language: Python 3.10

Python was chosen for its extensive library support, ease of use, and rapid prototyping capabilities.

Libraries and Frameworks:

FastAPI: Used to create a high-performance API layer for image upload and response handling.

EasyOCR: Applied for text detection and recognition in images.

PIL (Pillow): Utilized for image rendering, drawing bounding boxes, and overlaying text.

Requests: For interfacing with the MyMemory Translation API.

NumPy: Used for pixel-level image manipulation and numerical operations when needed.

Execution Environment:

The entire system was tested on a CPU-based local machine, making the solution hardware-friendly and accessible.

Platform:

The backend was deployed and tested locally using FastAPI, allowing for RESTful interactions via endpoints such as /upload, /translate, and /process.

This environment ensured a fast, interactive development cycle with reproducible results and performance benchmarking.

Results

a. Screen Shot of Simulator/Emulator/Hardware Kits

Manga Panel Translator

Upload File (.png, .jpg, .jpeg):

Choose File

gm.png

Source Language:

Japanese

Target Language:

English

Translate

Original Preview:

おはよう

Processed Image:

Good morning

Download Processed Image

Detected & Translated Texts:

Good morning (x: 47, y: 49)

b. Graphs and Tables

Metric	Value
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OCR Accuracy	~85%
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Translation Latency	~1 sec per block
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Font Size Range	10 - 40 px
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c. Comparative Chart

Feature	Proposed System	Google Translate App
OCR Language Support	JA/EN	Multiple
In-Image Overlay	Yes	Yes
Font Size Adjustment	Dynamic	Fixed
Custom Deployment	Yes (FastAPI)	No

References/Bibliography

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